SIMULATED
Asset Management

NAV and settlement monitoring

Agent: **Operations**

OPENING PROMPT

```
Show me all failed settlements from the past 3 business days and help me understand what's causing them.
```

The demo date is the latest price date in the Snowflake Public Data share at build time, and positions are month-end. Agents answer from live data, so figures, names and rankings in your environment can differ from the examples in this script.

Business Context Setup

Persona: Tom Hughes, Middle Office Operations Manager at Simulated Asset Management
Business Challenge: Middle office operations teams must monitor trade settlements, reconciliation breaks, NAV calculations, corporate actions, and cash management across multiple portfolios and custodians. Manual monitoring is time-consuming, error-prone, and fails to provide real-time exception alerts, risking operational failures, NAV errors, and settlement penalties.
Value Proposition: AI-powered operations intelligence that provides real-time monitoring of all middle office processes, automated exception detection and root cause analysis, and intelligent remediation recommendations—transforming reactive operational fire-fighting into proactive exception management.

Agent: `AM_operations_copilot`
Data Available: 11 portfolios, 8 custodians, daily settlement activity, reconciliation data, NAV calculations (struck for each position date; month-end positions advance with the weekly refresh, latest month-end at time of writing 2026-09-30), corporate actions, cash positions

***Note**: NAV and reconciliation are derived from positions, so they advance automatically with the weekly data refresh as each new month-end lands. No full rebuild is required.*

Demo Flow

Scene Setting: Tom needs to review the latest operational status across all middle office functions before the next NAV sign-off. He needs to quickly identify and resolve any settlement failures, reconciliation breaks, or issues that could impact the NAV strike.

Step 1: Settlement Failure Monitoring

User Input:

PROMPT

```
Show me all failed settlements from the past 3 business days and help me understand what's causing them.
```

Tools Used:

- `middle_office_analyzer` (Cortex Analyst) - Query FACT_TRADE_SETTLEMENT for Status='Failed' in the most recent business days from SAM_MIDDLE_OFFICE_VIEW

Expected Response:

- Table showing: Trade ID, Security, Portfolio, Counterparty/Custodian, Settlement Amount, Days Old, Failure Reason
- A handful of failures in the window, all populated with a failure reason, plus any **aged fail** just outside the window that breaches the escalation threshold
- Severity flags: FAILED (aged beyond the settlement cycle - critical), PENDING (within the settlement window). The cycle depends on market: US trades settle T+1 (since May 2024); UK and EU trades settle T+2 until they move to T+1 on 11 October 2027
- Note the realistic base rates across the settlement history: Settled ~94%, Failed ~4%, Pending ~2%. A handful of failures per day is the expected shape, not a crisis
- Root cause analysis grouped by the three failure reasons in the data, each pointing to a different owner:
  - **Insufficient shares** — a stock-record problem (unprocessed corporate action, an upstream unsettled leg); usually the largest by value and the one behind the aged fail
  - **SSI mismatch** — standing settlement instructions differ between SAM and the counterparty or custodian
  - **Counterparty system issue** — on the counterparty's side; chase and monitor
- A chart of failed value by failure reason
- Specific remediation steps with responsible parties
- Data timestamp: "As of DD MMM YYYY HH:MM"

Talking Points:

- **Real-Time Operations Monitoring**: Instant visibility into settlement status across all counterparties
- **Automated Root Cause Analysis**: AI identifies specific failure reasons (SSI mismatch, counterparty issues)
- **Severity Classification**: Critical vs pending settlements based on age and regulatory requirements
- **Actionable Remediation**: Specific steps to resolve each failure with clear ownership

Key Features Highlighted:

- **Operational Intelligence**: Real-time exception detection across all settlement activity
- **Root Cause Identification**: Automated analysis of failure reasons and remediation paths
- **Regulatory Awareness**: Market-aware settlement monitoring (US on T+1 since May 2024; UK and EU moving from T+2 to T+1 on 11 October 2027) with escalation for aged failures

Step 2: Reconciliation Break Investigation

Presenter Transition:

> *"Settlement failures are under control with clear remediation paths. Two of them were insufficient-shares fails, which is exactly the sort of thing that shows up as a reconciliation break. So let me move on to reconciliation and review the latest status across all portfolios..."*

Reasoning: Settlements and reconciliation are interdependent; a settlement failure often causes a reconciliation break. This step ensures we have visibility across all operational risks before NAV calculation.

The settlement answer does not usually mention reconciliation, so the presenter makes the link explicit rather than relying on the agent's closing offer.

User Input:

PROMPT

```
Summarize the latest reconciliation breaks for all portfolios and flag any that are
critical.
```

Tools Used:

- `middle_office_analyzer` (Cortex Analyst) - Query FACT_RECONCILIATION for Status='Open' breaks from SAM_MIDDLE_OFFICE_VIEW, ordered by Difference DESC

Expected Response:

- The latest reconciliation date's open breaks: a small number across a few portfolios, typically all at Investigating
- Table of breaks: Break Type, Portfolio, Security, Difference Amount, Status, Severity
- Break types are exactly three: **Position**, **Cash**, **Price** (Price is the most common across the history). Statuses are Open, Investigating, Resolved

- Severity classification:
  - CRITICAL: Position breaks >$1M or >1% of NAV
  - HIGH: Position breaks above the $100K zero-tolerance limit, or cash breaks >$100K or >0.1% of NAV
  - Medium/Low: smaller or zero-difference breaks (a zero-difference cash break is a timing marker)
- In the latest run no break was CRITICAL, but **position breaks on single names exceeded the $100K limit** and need escalation
- Root cause hypotheses for each break (unprocessed corporate action, trade booked to the wrong account, unsettled leg) and resolution steps
- Impact on NAV calculation if unresolved

Talking Points:

- **Intelligent Break Prioritisation**: AI automatically classifies breaks by severity and NAV impact
- **Root Cause Analysis**: Identifies whether breaks are due to corporate actions, timing, or system errors
- **NAV Impact Assessment**: Calculates potential NAV errors if breaks remain unresolved
- **Resolution Guidance**: Specific remediation steps with timeline to resolution

Key Features Highlighted:

- **Break Classification**: Automated severity assessment based on amount and type
- **Corporate Action Detection**: Links breaks to unprocessed corporate actions
- **NAV Protection**: Identifies breaks that could cause NAV calculation errors

Step 3: NAV Calculation Status

Presenter Transition:

> *"Reconciliation breaks are identified and prioritised. Ahead of the NAV sign-off deadline, let me check the latest NAV status across all funds and identify any anomalies that could indicate underlying issues we've missed..."*

Reasoning: NAV calculation is the culmination of all middle office processes. Checking NAV status validates whether settlement and reconciliation issues are impacting valuations.

User Input:

> **PROMPT**
>
> ```
> What's the status of the latest NAV calculation across all funds? Are there any anomalies I need to investigate?
> ```

Tools Used:

- `middle_office_analyzer` (Cortex Analyst) - Query FACT_NAV_CALCULATION for latest NAV by portfolio from SAM_MIDDLE_OFFICE_VIEW

Expected Response:

- NAV calculation summary: the latest NAV run is **complete for all 11 funds, every fund Calculated and Approved**
- Table by fund: Fund Name, NAV per share, Net Assets, Status, Anomalies Detected
- Total net assets across the 11 funds **ties to firm AUM** from holdings (both a little under $7.5bn at time of writing) — NAV reconciles to holdings, which is the point to make if a sceptical audience asks
- A small number of funds carry a recorded anomaly despite approval, typically a **Zero NAV Movement Anomaly** (NAV per share unchanged versus the prior day), which the agent reads as a possible stale-price issue
- Recommended checks: pricing completeness for the flagged funds, cross-check against the open reconciliation breaks from Step 2, then document the override or re-run with corrected prices
- The remaining funds are reported clean

Talking Points:

- **Automated Anomaly Detection**: AI flags anomalies even on approved NAVs, rather than taking approval as the end of the story
- **Root Cause Intelligence**: Links NAV anomalies to specific operational issues (stale prices, breaks, corporate actions)

- **Approval Workflow**: Clear recommendations on which NAVs can be relied on downstream and which need a same-day check

Key Features Highlighted:

- **NAV Anomaly Detection**: Automated identification of unusual valuation movements
- **Cross-Functional Analysis**: Links NAV issues to reconciliation breaks and corporate actions
- **Approval Intelligence**: Risk-based recommendations for NAV approval

Step 4: Corporate Action Processing Status

Presenter Transition:

> *"Every NAV is approved, with a couple of anomalies to check. But the best operations teams think ahead—corporate actions due this week could cause tomorrow's issues. Let me review what's coming up and ensure we're prepared..."*

Reasoning: Proactive operations management requires forward-looking visibility. Corporate action review prevents tomorrow's reconciliation breaks and NAV issues.

User Input:

> **PROMPT**
>
> ```
> Show me all pending corporate actions for the coming business days and highlight any that require immediate processing.
> ```

Tools Used:

- `middle_office_analyzer` (Cortex Analyst) - Query FACT_CORPORATE_ACTIONS for upcoming actions from SAM_MIDDLE_OFFICE_VIEW
- `search_internal_docs` (Cortex Search) - Get custodian notifications for pending corporate actions

Expected Response:

- A dozen or so unprocessed corporate actions (Pending or Announced) with ex-dates over the coming business days, all still awaiting processing
- Table: Security, Action Type, Ex-Date, Payment Date, Portfolios Affected, Status, Priority
- Priority tiers:
  - **Immediate**: a **merger** with ex-date today (pending regulatory approval: tag holdings and confirm election deadlines) and a **stock split** on a widely held name with ex-date the next business day (prepare the entitlement calculation now, or risk position reconciliation breaks across every portfolio holding it). In the latest run these were Mastercard and a Microsoft 2-for-1 split
  - **Monitor**: buybacks and a dividend in the next couple of days needing rate confirmation and accrual set-up
  - **Routine**: regular and special dividends and buybacks later in the week
- Processing recommendations for each action type: dividend accruals, split position updates, cash forecasting
- Coordination requirements (trading desk, portfolio accounting)

Talking Points:

- **Proactive Corporate Action Management**: Forward-looking view of all upcoming events
- **Priority-Based Alerts**: Automatic flagging of actions requiring immediate processing
- **Break Prevention**: The split is the clearest example of a corporate action that becomes tomorrow's reconciliation break if missed
- **Cross-Team Coordination**: Identifies which teams need to be involved for complex actions

Key Features Highlighted:

- **Forward-Looking Monitoring**: Forward corporate action calendar with priority classification
- **Processing Intelligence**: Automated recommendations for how to handle each action type
- **Impact Analysis**: Calculates NAV and cash impact of all pending actions

Scenario Wrap-up

Business Impact Summary:

- **Operational Efficiency**: Reduced manual monitoring time by 70% through automated exception detection

- **Risk Reduction**: Proactive identification of settlement failures, reconciliation breaks, and NAV anomalies before they become critical
- **NAV Accuracy**: Prevented NAV errors through automated anomaly detection and corporate action tracking
- **Regulatory Compliance**: Timely settlement monitoring supports compliance with shortening settlement cycles (US T+1 since May 2024; UK and EU T+1 from 11 October 2027) and avoids settlement penalties

Technical Differentiators:

- **Real-Time Operations Intelligence**: Continuous monitoring across all middle office functions (settlement, reconciliation, NAV, corporate actions, cash)
- **Automated Root Cause Analysis**: AI-powered investigation of failures, breaks, and anomalies with specific remediation recommendations
- **Cross-Functional Integration**: Links operational issues across settlement, reconciliation, and NAV calculations for comprehensive root cause identification
- **Severity-Based Prioritisation**: Intelligent classification of exceptions by business impact and regulatory urgency